

Project Proposal Instructions

1. Objective of the Proposal

The project proposal is intended to describe what project your group plans to conduct, which generative models will be used, and how the project will be carried out.

The proposal should demonstrate:

- Understanding of generative models covered in the course
- Ability to choose appropriate models and datasets
- Clear planning and feasibility within the course timeline

The proposal is a planning document, not a final report.

2. Proposal Length & Format (Strict Requirements)

Length

- **1–2 pages maximum** (excluding references)

Formatting

- Font: Times New Roman (or equivalent)
- Font size: 11 or 12
- Line spacing: 1.15 or 1.5
- Margins: Standard (2–2.5 cm)
- Text alignment: Justified
- File format: **PDF only**

Proposals that do not follow the formatting requirements may be penalized.

3. Team Requirements

- Team size: **1–3 students**
- All team members must be listed on the first page
- Individual roles or responsibilities should be briefly stated

4. Required Structure of the Proposal

The proposal *must follow the structure below*. All sections are mandatory.

4.1 Title Page

The first page should include:

- Project title
- Course name: *Generative Models*
- Names of all team members
- Student IDs
- Team size

4.2 Problem Statement & Motivation

This section should explain:

- What problem the project addresses
- Why this problem is interesting or relevant
- Why a generative model is appropriate for this task

The motivation should clearly connect the project to the course content.

4.3 Background & Related Work (High-Level)

Provide a brief overview of:

- Existing work related to your task
- Relevant generative models discussed in the course
- High-level ideas behind the chosen models

Detailed mathematical derivations are not required.

4.4 Chosen Generative Models & Algorithms

Clearly describe:

- Which generative model(s) will be used (e.g., VAE, GAN, Diffusion, Transformer, LLM)
- Why these models were chosen
- Whether the project involves:
 - Training from scratch
 - Fine-tuning a pre-trained model
 - Using a pre-trained model for analysis or evaluation

4.5 Dataset Description

Specify:

- Dataset name(s)
- Type of data (image, text, audio, multimodal, etc.)
- Approximate dataset size
- Source of the dataset (public benchmark, online repository, etc.)

If training or fine-tuning is involved:

- Briefly describe preprocessing steps

4.6 Methodology & Experimental Plan

Describe:

- Overall workflow of the project
- Planned training or fine-tuning strategy
- Tools and frameworks to be used (e.g., PyTorch, TensorFlow)

This section should focus on what will be done, not implementation details.

4.7 Evaluation Strategy

Explain how the project results will be evaluated:

- Qualitative evaluation (e.g., visual inspection, generated samples)
- Quantitative metrics (if applicable)
- Comparisons between models or configurations (if any)

4.8 Expected Outcomes & Challenges

Discuss:

- Expected results or deliverables
- What insights the group expects to gain
- Potential challenges or limitations

4.9 Project Timeline

Provide a brief timeline including:

- Dataset preparation
- Model selection or implementation
- Training and experimentation
- Evaluation
- Final report preparation

A simple table or bullet list is sufficient.

4.10 References

- Include all referenced papers, articles, datasets, and documentation
- Use a consistent citation style

- Online resources are acceptable

5. Scope & Expectations

- Projects must be related to generative models covered in the course
- The project does not need to be state-of-the-art
- Emphasis is on understanding, clarity, and feasibility

6. Evaluation Criteria for the Proposal

Proposals will be assessed based on:

- Clarity and organization
- Relevance to generative models
- Appropriateness of chosen models and datasets
- Feasibility within course constraints
- Demonstrated understanding of course material

7. Important Notes

- Plagiarism is not allowed
- All team members are responsible for the proposal content
- Ethical considerations (e.g., data bias, misuse of generated content) should be acknowledged when relevant